

Current Management of Cerebral Venous Thrombosis

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ABSTRACT: Cerebral venous thrombosis is a rare stroke type that primarily affects younger women. Contemporary large international efforts have improved our understanding of the natural history and management of this rare disease, yet important evidence gaps persist across the disease continuum. Management of cerebral venous thrombosis can be conceptualized into multiple phases: acute management, primary therapy, secondary prevention, and chronic recovery. Acute treatment centers on anticoagulation, including in the presence of intracranial hemorrhage. Endovascular therapy and decompressive craniectomy are reserved for selected severe cases. Direct oral anticoagulants appear comparable to vitamin K antagonists for most patients during primary treatment, though optimal lead-in duration and treatment length remain uncertain. Decisions regarding extended anticoagulation for secondary prevention require individualized assessment of recurrence and bleeding risk. Patients with cancer, antiphospholipid antibody syndrome, prior venous thromboembolism, and idiopathic events are at the highest thromboembolic risk. Long-term sequelae, including fatigue, headache, cognitive and mood disturbances, epilepsy, and intracranial hypertension, contribute substantially to morbidity despite high rates of functional independence. Heavy menstrual bleeding may impact young women on anticoagulation. Recognition and management of these outcomes are essential for comprehensive care.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: brain ■ epilepsy ■ fatigue ■ intracranial hypertension ■ thrombosis

CURRENT MANAGEMENT OF CEREBRAL VENOUS THROMBOSIS

Overview

Cerebral venous thrombosis (CVT) comprises 0.5% to 3% of stroke¹ and has an incidence of 10 to 20/million per year. DVT/PE is $\approx 100\times$ more common.² Certain populations are at increased risk, however, particularly younger women due to associations with oral contraceptive (OCP) use and pregnancy. Outside of reproductive years, incidence, outcomes, and risk factors become similar between men and women.^{3–7}

As a venous thromboembolic event affecting the brain, CVT occupies the intersection between stroke neurology and thrombosis medicine. Multidisciplinary coordination is often required to optimize outcomes.

Newer observational studies and randomized trials have informed recent guidelines and societal statements.^{1,8,9} This review synthesizes current management approaches, emphasizing individualized care, ongoing controversies, and research priorities. Trauma-associated, pediatric, incidental, and vaccine-induced thrombotic thrombocytopenia-related CVT are beyond the scope of this review and have been recently addressed elsewhere.^{8,10–12}

Diagnosis

The severity of clinical presentations can range from isolated headache to coma or status epilepticus. Symptoms are influenced by patient factors as well as thrombus burden and location, associated parenchymal injury or

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hemorrhage, intracranial pressure (ICP) or mass effect, and seizure burden.

Unlike an arterial stroke, symptom onset is frequently subacute. Fewer than half of patients present within 48 hours, and diagnostic delay is common because symptoms can be nonspecific.^{13–15} A minority of patients will present hyperacutely with thunderclap headache, sudden-onset focal deficits, or seizures. Headache occurs in over 90%, but less commonly with isolated cortical vein thrombosis and in older adults.¹⁶ Other common manifestations include focal deficits, seizures, vision loss, cranial neuropathies, and encephalopathy.¹

Up to 40% of patients have multiple health care encounters before diagnosis, particularly younger women.^{17,18} Early recognition improves patient experience and health-system efficiency. Delayed diagnosis has not consistently been associated with worse outcomes, but this may be due in part to CVT's favorable natural history, and because those presenting early with severe syndromes may be more thoroughly investigated.^{19–21}

Laboratory Investigations

Unlike for suspected DVT/PE, D-dimer is insufficiently sensitive to distinguish individuals with suspected CVT who require vascular neuroimaging, particularly with less severe presentations (isolated headache, chronic presentations, low clot burden). Emerging multimodal and machine-learning approaches integrating noncontrast neuroimaging, laboratory investigations, and clinical information may improve risk stratification and diagnosis.²²

Imaging

Noncontrast CT findings prompting suspicion for CVT include hyperdense venous sinuses or cortical veins, parenchymal edema, and intracerebral hemorrhage (ICH) in a nonarterial distribution, or juxtacortical hemorrhages. However, noncontrast CVT is interpreted as normal in up to two-thirds of patients.²³ CT venography or MR venography remains the diagnostic standard. Magnetic resonance imaging allows detailed assessment of parenchymal lesions, and susceptibility-weighted sequences improve the detection of isolated cortical vein thrombosis.¹ Black-blood thrombus imaging, a magnetic resonance imaging technique directly visualizing intraluminal thrombus, demonstrates high diagnostic accuracy in single-center studies but has not yet been validated against standard venographic imaging.²⁴

Early Clinical and Neuroimaging Predictors of Prognosis

Several clinical-radiological prognostic scores for CVT exist but have moderate external validity.^{25–27} Baseline clinical factors most consistently linked to in-hospital

mortality and functional dependence include older age, reduced consciousness, and comorbid malignancy or multimorbidity.^{4,25–29} Beyond the acute period, older age and malignancy are associated with 1-year mortality.^{3,29} Neuroimaging features predictive of worse outcomes include parenchymal lesions with mass effect or ICH.

Parenchymal lesions occur in up to two-thirds of patients and correlate with elevated venous pressure, though patterns vary, likely from anatomic variations in venous collateralization.^{30–33}

Increased venous hydrostatic pressure likely contributes to blood–brain barrier disruption, leading to vasogenic edema and, in severe cases, hemorrhage and cytotoxic injury. However, the relative contributions of hydrostatic and inflammatory mechanisms remain unclear. Early venous recanalization is associated with fewer new or progressive nonhemorrhagic lesions but does not appear to reduce established ICH.³⁴ Acute restricting diffusion of parenchymal lesions does not appear to reliably predict persisting parenchymal injury at day 90.³⁴ However, baseline ICH, which is present in 30% to 40% of cases, is associated with worse chronic outcomes, including functional dependence and epilepsy.^{25,35} Whether this reflects severe primary injury or secondary injury from blood products is uncertain.

The prognostic value of other imaging modalities employed in other stroke types, including perfusion, tractography, volumetrics, and functional magnetic resonance imaging, remains underexplored in CVT and warrants further study.

ACUTE MEDICAL MANAGEMENT

Following vital stabilization, goals of acute management include reducing thrombus burden and preventing further parenchymal injury, treating the underlying hypercoagulable state, and controlling complications including headache, seizures, and increased ICP.

Anticoagulation

Acute Anticoagulation

Acute anticoagulation aims to prevent thrombus propagation, facilitate recanalization, and treat systemic hypercoagulability. Guidelines recommend initial parenteral therapy, with subsequent transition to oral anticoagulation.^{1,36} Low-molecular-weight heparin is generally preferred over unfractionated heparin because of more predictable pharmacokinetics and lower risk of heparin-induced thrombocytopenia. Unfractionated heparin is reserved for severe renal impairment or when rapid reversal may be required.^{1,8} Heparin-induced thrombocytopenia–related CVT, though rare, should be considered in patients with thrombocytopenia or new thrombosis during heparin exposure, and requires non-heparin anticoagulation.³⁷

In a prospective neuroimaging study, recanalization with anticoagulation was common within the first 8 days of therapy; 68% achieved partial, and 6% complete, recanalization.³⁴ Anticoagulation is recommended even in the presence of hemorrhage: reduction of thrombus burden and recanalization may limit venous pressure and thus prevent further bleeding. Although late hemorrhage (new or expanding hemorrhagic lesions following initiation of anticoagulation) is reported in 6% to 14% in case series, data are limited, and the incidence of symptomatic cases is not known.^{34,38–40}

In recent large observational series, 10% to 12% received DOAC for acute anticoagulation.^{41,42} Data is currently insufficient to confirm the safety or efficacy of forgoing initial parenteral therapy, or to identify optimal candidates for DOAC-only management.^{41,42}

Supportive Care

Seizures may impact over one-third of patients during admission. Risk factors include parenchymal lesions, cortical vein or sagittal sinus involvement, and female sex-specific risk factors.^{43,44} Current evidence does not support antiseizure prophylaxis in the absence of seizures.¹¹ Intracranial hypertension is common, with papilledema present in over one-quarter of cases.^{13,31} Multiple dural sinus involvement, particularly the combination of superior sagittal and transverse sinuses, is a risk factor.⁴⁵ ICP and papilledema typically improve with venous recanalization, but acetazolamide may be used in patients with ICP-related visual symptoms or papilledema.¹¹ Because visual loss can occur without overt symptoms, early ophthalmology involvement is ideal to assist in diagnosing and monitoring papilledema. A headache may initially be severe. Management includes analgesia, measures to reduce increased ICP, and anticoagulation to facilitate recanalization. Beyond the reduction of increased ICP, the role of acetazolamide for headache management remains uncertain.⁴⁶

ACUTE ENDOVASCULAR MANAGEMENT

Endovascular therapy (EVT) has transformed outcomes in large-vessel occlusion stroke, but comparable evidence for CVT is lacking. Given that most patients with CVT achieve good functional outcomes with medical management alone, the optimal population for EVT remains uncertain, and it is estimated that only 1% to 5% of patients with CVT undergo EVT.^{47–49} Guidelines currently recommend EVT as rescue therapy for patients who deteriorate despite anticoagulation or have contraindications to it. Identifying predictors of early neurological deterioration (currently without a standardized definition) may facilitate selection for EVT or other emerging acute therapies and remains a priority for CVT research.⁵⁰ Observational data suggest better functional outcomes

in patients who are treated early (<24 hours) compared with later with EVT, though optimal timing and candidate selection remain unclear.⁵¹

No trial has evaluated EVT as a first-line therapy alongside anticoagulation. In the only randomized trial to date, TO-ACT, EVT did not improve 6- or 12-month functional outcomes despite higher recanalization rates.⁵² The trial was stopped early for futility and was limited by broad eligibility criteria for high-risk features, heterogeneous EVT techniques, and procedural complications. The 9% incidence of sinus perforation in TO-ACT exceeds the 1.1% rate reported in pooled analyses, including more recent studies using newer-generation devices.⁵³ Subsequent studies have reported a 5% to 9% risk of ICH.⁵³ Meta-analyses of observational studies report high rates of favorable outcomes in EVT-treated patients but likely reflect selection bias.^{53,54} The absence of standardized technical approaches or outcome measures further complicates the interpretation of results.⁵⁵

The primary goal of EVT in CVT is to reduce clot burden and restore venous outflow, thereby improving the efficacy of anticoagulants or thrombolytic, but technical challenges arise from sinus anatomy and the lack of optimized devices.⁵⁶ A recent meta-analysis identified common techniques including aspiration thrombectomy (~30%) and combined aspiration–stent retriever techniques (~20%), often with adjunctive thrombolysis (~50%). Although there is no accepted classification for angiographic recanalization, aspiration techniques appear to yield the highest proportion of favorable outcomes, without significant differences between methods for mortality or complication rates. Notably, reported outcomes with local thrombolysis alone are similar to those achieved with mechanical thrombectomy.⁵⁷

Most current devices are designed for arterial use and are suboptimal for large-caliber, thin-walled venous sinuses. New systems, such as large-bore aspiration catheters and modified 3-dimensional stents,^{58,59} aim to address these limitations.⁶⁰ Cortical vein thrombosis in particular remains a technical challenge.

α -2 antiplasmin inhibitors, which enhance fibrinolysis, are being evaluated in phase II trials for acute ischemic stroke and DVT/PE.²² Pending results, they may represent a future area of investigation as an adjunct or alternative to anticoagulation or EVT to promote early recanalization.²²

DECOMPRESSIVE CRANIECTOMY

Decompressive craniectomy is reserved for patients with severe mass effect with midline shift or impending herniation. In DECOMPRESS-2, the largest prospective study to date, mortality and functional dependence at 12 months reached 64%, exceeding rates in earlier case series.^{61,62} Seizures occurred in 23% at 1 year.⁶³ As in ischemic stroke, the benefits of decompressive

craniectomy appear greatest with early intervention before marked deterioration; outcomes were worse in patients with preoperative coma or fixed pupils.

POSTACUTE MANAGEMENT

Anticoagulation and Secondary Prevention After CVT

Primary Anticoagulation (Initial 3–12 Months)

Following medical stabilization, patients transition to oral therapy. Options for most individuals include a DOAC or vitamin K antagonist. The optimal duration of initial parenteral therapy before transition to DOAC remains uncertain. Some trials required 5 to 15 days of parenteral anticoagulation, whereas others permitted shorter lead-in periods without clear safety concerns.^{30,64,65} Available data suggest that shorter durations of parenteral therapy are associated with milder clinical presentations, currently precluding conclusions regarding the relative safety and efficacy of shorter versus longer durations.^{42,66}

Evidence from small randomized trials and large observational cohorts suggests comparable safety and efficacy of DOACs and vitamin K antagonists for most CVT patients.^{14,30,41,64,65,67} However, several populations remain underrepresented, including those with trauma-associated CVT, pregnancy, antiphospholipid antibody syndrome, active cancer, recurrent venous thromboembolism (VTE), or severe presentations (Table S1). In the pediatric EINSTEIN-Jr trial, infection-associated CVT accounted for two-thirds of cases, demonstrated reassuring safety of DOACs in this population.⁶⁸

The optimal duration of primary anticoagulation remains uncertain. Guidelines recommend 3 to 12 months for patients without an indication for permanent anticoagulation.^{1,8,36} In clinical practice, duration within the 3 to 12 month timeframe is variable.^{14,30,69} The EXCOA-CVT cluster randomized trial compared sites using 3 to 6 versus 12 months of anticoagulation. Interim analyses found no differences in recurrent VTE, bleeding, or mortality, although 2-year results are pending.^{70,71}

It is controversial as to whether recanalization should guide the duration of primary anticoagulation. Most recanalization appears to occur within the first 3 months, with limited change thereafter.^{30,34,72} It is also unclear whether chronic recanalization independently improves outcomes. A meta-analysis of mostly retrospective data found that partial or complete recanalization is associated with lower odds of functional dependence, recurrence, and chronic headache.⁷³ It is uncertain, however, whether the benefits of recanalization are primarily due to reduced parenchymal injury within the very early period following initiation of anticoagulation.^{34,74}

Extended Anticoagulation and Secondary Prevention

After completion of primary anticoagulation, clinicians must determine whether to discontinue therapy or continue anticoagulation for secondary prevention. This decision requires balancing recurrence risk against bleeding risk and incorporating patient preferences through shared decision-making.

Indefinite anticoagulation is indicated for patients with persistent high-risk conditions such as active cancer, antiphospholipid antibody syndrome, severe inherited thrombophilia, or recurrent VTE. In contrast, anticoagulation can generally be discontinued in patients whose CVT was clearly provoked by a transient risk factor, particularly estrogen use or pregnancy.

In DOAC-eligible patients who are candidates for extended therapy, defining the optimal secondary prevention secondary prevention regimen remains an important research priority.⁵⁰

Recent VTE trials suggest that extended low-dose apixaban or rivaroxaban may offer safe and efficacious prophylaxis compared with full-dose therapy in both high- and intermediate-risk patients.^{75–78} The role of aspirin for VTE prophylaxis after CVT is uncertain. In general VTE populations, aspirin reduced recurrence by approximately one-third without a significant increase in bleeding over 3 years of follow-up.⁷⁹ In the EINSTEIN-CHOICE trial, however, aspirin was found to be inferior to low-dose rivaroxaban for recurrent VTE, without significant differences in bleeding over 1 year of follow-up.⁷⁸ Applying these findings to CVT requires caution, given the younger age and distinct bleeding risk profile of this population, particularly the potential for heavy menstrual bleeding.

Factor XIa inhibitors are novel anticoagulants under investigation for DVT/PE, stroke prevention in atrial fibrillation, and secondary stroke prevention. In the recent OCEANIC trial, asundexian, an oral FXIa inhibitor, reduced recurrent ischemic stroke without increasing bleeding in antiplatelet-treated patients.⁸⁰ Whether this class has a role in CVT remains unknown but may warrant evaluation in future trials.

Risk Stratification for Recurrence

The overall risk of recurrent VTE after CVT is $\approx 4\%$ to 7% , with about half representing recurrent CVT.^{81,82} Available data are limited by relatively small cohorts and short follow-up. Risk factors for recurrence parallel those seen in the DVT/PE population, with highest rates observed in patients with active cancer, antiphospholipid antibody syndrome, prior VTE, particularly if unprovoked.^{81,83,84} In some series, unprovoked CVT also carries a higher risk, particularly among women without identifiable triggers.^{14,81,85} In ACTION-CVT, antiphospholipid antibody positivity and Black race were also associated with recurrence, the latter likely reflecting disparities in social determinants of health.^{81,85}

Conversely, recurrence risk is lowest in provoked CVT, especially in women whose events occurred during estrogen use or pregnancy/puerperium. In a cohort of >200 patients without major thrombophilia (median follow-up 3 years), 3-quarters were women with sex-specific risk factors, mostly OCP use.⁸⁶ Even in this low-risk group, recurrence rates were higher in men and in factor V Leiden heterozygotes, while prothrombin gene heterozygotes had a numerically but nonsignificantly increased risk.

Thrombophilia and Malignancy Screening

Recent ASH guidelines recommend venous thrombophilia testing for patients considering cessation of anticoagulation.⁹ However, numbers needed to screen to prevent recurrent events with anticoagulation were derived from studies including individuals with known high-risk thrombophilias. The overall role of screening remains debated, but potential advantages include improved risk stratification and reassurance for low-risk patients concerned about recurrence. Screening for paroxysmal nocturnal hemoglobinuria may be considered in patients under 50 with suggestive features such as cytopenias or hemolysis, in consultation with a thrombosis specialist.⁸⁷ Routine cancer screening is not recommended for all patients with CVT. Absolute rates of subsequent cancer diagnosis are highest in patients over 50, particularly men.^{88,89} Of note, many patients with subsequent diagnosis of myeloproliferative disorder do not meet criteria at the time of their CVT diagnosis and later develop elevated cell counts.^{90–92} Younger adults have a higher relative rate of incident cancer following CVT compared with age-matched controls;^{88,89} thus they may benefit from education and follow-up for signs and symptoms concerning cancer.

Special populations

Certain populations require tailored antithrombotic strategies. In antiphospholipid antibody syndrome—particularly triple-positive patients or those with prior arterial events—vitamin K antagonists are preferred, as DOACs are associated with higher rates of breakthrough thrombosis in this group.^{93–96} DOACs are contraindicated during pregnancy and breastfeeding, and vitamin Ks are teratogenic. Current VTE guidelines support the use of factor Xa inhibitor DOACs, particularly apixaban, as an alternative to LMWH in patients with active cancer, although CVT-specific data are lacking.^{97–100} Breakthrough CVT during DOAC therapy should prompt evaluation for underlying causes, such as malignancy or antiphospholipid syndrome; transition to LMWH is common. Collaboration with thrombosis specialists is advisable for complex cases.

Sex-Specific and Reproductive Considerations

Patients who menstruate should be counseled regarding heavy menstrual bleeding (HMB, also termed abnormal

uterine bleeding), which affects up to 70% receiving anticoagulation, and be monitored for iron deficiency and anemia. HMB can impair quality of life and reduce adherence; management strategies have been reviewed recently.^{101,102}

Some patients may have preexisting HMB that was managed with OCPs, and may worsen if OCPs are discontinued. Evidence from the VTE literature suggests OCP continuation during anticoagulation is safe.¹⁰³ Because estrogen-related prothrombotic effects may persist for several weeks, OCP should be stopped at least 1 month in advance of a discontinuation of anticoagulation.¹⁰⁴

Women of childbearing potential should receive counseling regarding contraception, VTE risk, and LMWH prophylaxis for future pregnancies. Pregnancy after CVT is associated with a higher relative risk of recurrence compared with the general pregnant population (risk of CVT is $\approx 9/1000$ pregnancies, 16 $\times$ that of the general population; overall VTE risk is 27/1000, 80 $\times$ that of the general population), but absolute risk remains low and should not preclude future pregnancy.¹⁰⁵ For women who have discontinued anticoagulation, prophylactic low-dose LMWH during pregnancy is recommended, ideally with specialist obstetric involvement.¹⁰⁶

CHRONIC COMPLICATIONS

Persisting Nonmotor Symptoms

Although outcomes after CVT are often described as favorable, traditional measures such as the modified Rankin Scale underestimate the burden of nonmotor sequelae, including headache, fatigue, cognitive impairment, and mood disturbance. In the SECRET trial, most patients were functionally independent at baseline, yet impairments in cognition, mood, fatigue, and quality of life were common.³⁰ Symptoms generally improved over 12 months, particularly within the first 90 days, though other cohorts report persistent deficits affecting social and occupational participation and quality of life. Recognition, follow-up, and referral for supportive care, including peer-support resources, are essential. Further research is needed to identify predictors of adverse trajectories and effective management strategies.

Epilepsy

Approximately 10% of patients develop epilepsy. The strongest predictors include status epilepticus and decompressive craniectomy, followed by early seizures and intracranial hemorrhage.³⁵ The DIAS-3 risk score, incorporating these components, showed moderate predictive ability and may inform decisions regarding antiseizure medication discontinuation. Epilepsy is not adequately captured by global disability scales and

should be systematically assessed in future studies of CVT.

ICP and Vision

Intracranial hypertension persists chronically in about 10%. Risk factors include large clot volume and involvement of the sagittal or dominant transverse sinus.¹⁰⁷ Ongoing ophthalmology follow-up is important; Imaging findings such as optic nerve sheath enlargement may also assist in monitoring.^{107,108} In rare instances, persistent intracranial hypertension can occur despite venous recanalization.¹¹ The pathophysiology is not known and may be due to multiple factors, including arachnoid granulation scarring, transverse sinus stenosis, or shared risk factors for idiopathic intracranial hypertension.

Dural Arteriovenous Fistula

Dural arteriovenous fistula (dAVF) has a bidirectional association with CVT, where it can be a chronic complication of CVT in $\approx 2\%$ to 5% , or CVT can form adjacent to a preexisting dAVF. Many are diagnosed concurrently. Symptoms include new or changing headache, pulsatile tinnitus, focal deficits, or seizures. Risk factors for post-CVT dAVF include male sex, transverse sinus involvement, absent recanalization, and delayed diagnosis.¹⁵ Persistent venous occlusion likely contributes to the pathophysiology of dAVF through the presence of venous hypertension and subsequent neovascularization.¹⁰⁹ The literature does not support screening for dAVF in the absence of clinical suspicion; a substudy of the RE-SPECT CVT trial found that none of 120 participants developed dAVF at 180 days.¹¹⁰ The estimated median onset time from CVT onset, followed by dAVF diagnosis, is variable, ranging from 44 to 434 days and likely biased in part by duration of follow-up in CVT-based cohorts versus retrospective cohorts of people with dAVF.^{15,111,112} Given diagnostic challenges, detection may be improved with recommendations regarding imaging modalities and imaging review from experts in vascular neuroimaging.

SUMMARY

Management of CVT has evolved over the last decade. Large international efforts have refined the risks of chronic complications and improved the understanding of long-term prognosis. Multiple studies have clarified the role of contemporary therapies, including DOACs and decompressive craniectomy, although the role of EVT remains controversial. Future priorities for patients and providers include improved early diagnosis, identifying ideal candidates and approaches for acute

revascularization, and reducing the burden and duration of persisting symptoms.

ARTICLE INFORMATION

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Supplemental Material

Table S1



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